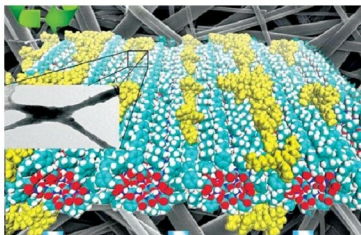


TECHNOLOGY AT YOUR SERVICE

Researchers develop biodegradable displays for electronics

Researchers at University of Missouri, USA, are on the verge of creating biodegradable electronics by using organic components in screen displays, which could help reduce electronic waste in the world's landfills. According to Suchismita Guha, professor in Department of Physics and Astronomy at MU College of Arts and Science, "This discovery creates the first biodegradable active layer in organic electronics, meaning, in principle we can eventually achieve full biodegradability."



A theoretical simulation of the distribution of the polymer on peptide nanotube and an electron microscopy image of the nanocomposite (Image courtesy: www.phys.org)

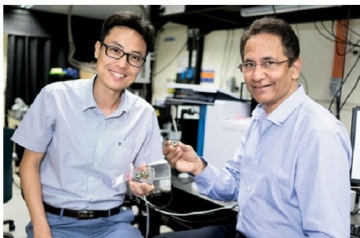
Guha, along with graduate student Soma Khanra, collaborated with a team from Federal University of ABC (UFABC) in Brazil to develop organic structures that could be used to light handheld device screens.

Using peptides, or proteins, they were able to demonstrate that these tiny structures when combined with a blue light-emitting polymer could successfully be used in displays. The scientists also discovered that by using peptide nanostructures, they could use fewer polymers. Using less to create the same blue light means that nanocomposites achieve almost 85 per cent biodegradability.

Scientists develop super-sensitive magnetic sensor

A hybrid magnetic sensor that is over 200 times more sensitive than most commercially-available sensors has been developed by scientists. This paves the way for the development of smaller and cheaper sensors for various fields such as consumer electronics, information and communication technology, biotechnology and automotive.

When an external magnetic field is applied to certain materials, a change in electrical resistance called



Researchers from National University of Singapore (Image courtesy: www.phys.org)

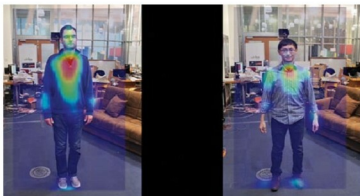
magnetoresistance, occurs as electrons are deflected.

The sensor developed by the team led by associate professor Yang Hyunsoo from National University of Singapore, is made of graphene and boron nitride, and comprises a few layers of carrier-moving channels, each of which can be controlled by the magnetic field. The scientists characterised the sensor by testing it at various temperatures, angles of magnetic field and with a different pairing material.

MIT researchers use Wi-Fi signals to capture human figure through walls

In 2012, researchers at University College in London used a modified Wi-Fi router to detect people through walls. Now, in 2015, researchers at Massachusetts Institute of Technology's (MIT's) Computer Science and Artificial Intelligence Lab have taken the concept forward and developed RF-Capture that not only detects human beings across walls but is also capable of recreating silhouettes from which it would be possible to distinguish people in a room based on their physical characteristics.

RF-Capture has been developed to create a rough image of the individual across the wall and even monitor



RF-Capture transmits wireless signals and then analyses those reflections to piece together a human form (Image courtesy: www.theverge.com)